

Separating Mixtures — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, name the **property that differs** between the substances — soluble or insoluble, boiling point, magnetic, particle size — because that property is why the method works.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe mixtures or experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- Naming a method scores one mark. Saying why it works is where the rest are.

SECTION 1 – PLANNING A FULL SEPARATION

- 1.** A dry mixture contains **iron filings, sand and salt**. Plan a method to obtain all three separately and dry. Give the steps in order, with a reason for each, and say what you have after every step. [7]

SECTION 2 – R_f IN REVERSE

- 2.** A dye has an R_f value of **0.80** in a particular solvent.
- (a)** On a chromatogram where the solvent front moves 9.0 cm, calculate how far the dye will travel. [2]

- (b)** A second spot on the same paper is 3.6 cm from the baseline. Calculate its R_f value. [2]

- (c)** The same dye is run in a different solvent and its R_f changes. Explain why the R_f can still be used to identify it. [1]

SECTION 3 – THE LONG EXPLANATION

- 3.** Describe and explain how simple distillation produces pure water from sea water. [6]
Your answer should cover what happens to the water and to the salt in the flask, what the condenser does and how, what the thermometer shows, and why the collected liquid is pure.

- 4.** Sea water is left in a shallow dish in the sun and, after several days, only white crystals remain. State the name of this process, what has happened to the water, and why the salt did not leave with it. [3]

5. A student says: "Chromatography could be used to get the water out of ink." [3]
Explain why this is wrong, and state what chromatography **would** tell you about the ink.

6. Two students wrote the following in a test. Neither answer is correct. [6]

(i) "To get the salt out of salt water, filter it — the salt stays on the paper."

(ii) " $R_f = \text{distance moved by the solvent} \div \text{distance moved by the spot}$."

For each one, explain the mistake and write a correct version.

End of paper. Check every answer has a unit or a form the question actually asked for.